

~~Guest~~
Video ML, DL, NLP
course
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Sec-3 Naive Bayes's
Theorem

lec 1 :- Understanding
Bayes's Theorem

lec 2 :- Variants of
Bayes's Theorem

Naive Bayes's Theorem
(Classification)

- ① Probability
- ② Bayes's Theorem

Independent Events

~~Rolling a~~ Rolling a
dice

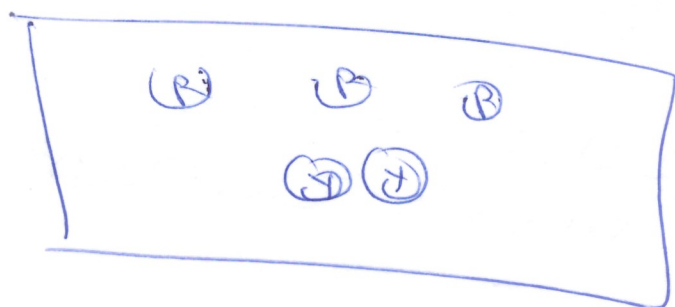
$\rightarrow \{1, 2, 3, 4, 5, 6\}$

Probability of 1 = $\frac{1}{6}$

Probability of 2 = $\frac{1}{6}$

Probability of 3 = $\frac{1}{6}$

Dependent Events

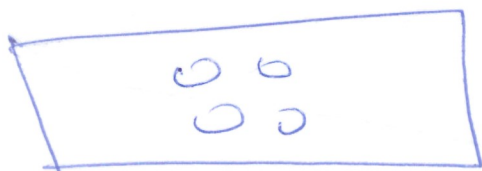


① What is the probability
of ~~orange~~ ~~white~~ ~~orange~~ remaining is
blue or yellow
more?

→ Probability (remaining
an orange
marble)
 $= \frac{3}{5}$

(3)

After removing an orange ~~the~~ marble
→ total 4 marbles left.



→ orange marble removed

Now,
probability of remaining
a yellow marble

$$P(+1) = \frac{3}{4} = \frac{1}{2}$$

$= \frac{1}{2}$
(2nd event)

Remove the yellow marble

$$P(0 \text{ and } +1) = P(0) * P(+1)$$

Probability of orange

Probability of yellow
given orange has been removed

$$P(0 \text{ and } 4) = P(0) * P(4|0)$$

→ we essentially compare them

This is known as conditional probability

$$= \frac{3}{5} * \frac{1}{2} = \frac{3}{10}$$

$$P(A \text{ and } B) = P(A) * P(B|A)$$

↙
This to be used for deriving Bayes theorem

Bayes's Theorem

$$P(A \text{ and } B) = P(B \text{ and } A)$$

$$P(A) * P(B|A) = P(B) * P(A|B)$$

$$P(A|B) = \frac{P(A) * P(B|A)}{P(B)}$$

Bayes's theorem ↙

$$P(A|B) = \frac{P(A) * P(B|A)}{P(B)}$$

(5)



Baye's Theorem.

$P(A|B)$ = Probability of Event A given B has occurred.

$P(A)$ = Probability of Event A

$P(B)$ = Probability of Event B

$P(B|A)$ = Probability of Event B given A has occurred.

~~Input~~

Independent Features			Dependent Features
x_1	x_2	x_3	Y
—	—	—	yes
—	—	—	No
—	—	—	yes
—	—	—	No

Now, we have to see how Bayes' Theorem is applied in ML for prediction

Suppose we calculate

$$P(+ | (x_1, x_2, x_3)) = \frac{P(+)*P(x_1, x_2, x_3 | +)}{P(x_1, x_2, x_3)}$$



Now expanding this equation as:-

$$\Rightarrow \frac{P(+)*P(x_1 | +)*P(x_2 | +)*P(x_3 | +)}{P(x_1)*P(x_2)*P(x_3)}$$

$$P(- | (x_1, x_2, x_3))$$

$$\Rightarrow \frac{P(-)*P(x_1 | -)*P(x_2 | -)*P(x_3 | -)}{P(x_1)*P(x_2)*P(x_3)} = 0.60 = 60\%$$

$$P(N_0 | x_1, x_2, x_3)$$

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$$= \frac{P(N_0) * P(x_1 | N_0) * P(x_2 | N_0) * P(x_3 | N_0)}{P(x_1) * P(x_2) * P(x_3)}$$

$$= 0.40$$

$$(1 - 0.60)$$

$\Rightarrow$ Let's solve a Normal Bayes theorem

Out look

	Yes	No	$P(E/Yes)$	$P(E/No)$
Sunny	2	3	2/9	3/5
Overcast	4	0	4/9	0/5
Rain	3	3	3/9	2/5

(8)

temperature

	yes	no	$P(E yes)$	$P(E no)$
Hot	2	2	2 2/9	2/5
Mild	4	2	2 4/9	4/5
Cool	3	1	1 3/9	3/5

Days (+ (N))

	1	$P(+ yes)$	$P(+ no)$
yes	9	9/14	5/14
no	5		

(9)

$$P(\text{tes} | \text{Sunny, Hot})$$

$$= \frac{P(\text{tes}) * P(\text{Sunny} | \text{tes}) * P(\text{Hot} | \text{tes})}{\cancel{P(\text{Sunny})} * P(\text{Hot})}$$


 2nd is constant value
 for tes & the both
 → hence its removed

$$\frac{9}{10} \times \frac{2}{9} \times \frac{2}{9}$$

$$= \frac{2}{03} = 0.031$$

$$P(\text{No} | \text{Sunny, Hot})$$

$$= P(\text{No}) * P(\text{Sunny} | \text{No}) * P(\text{Hot} | \text{No})$$

$$= \frac{8}{10} * \frac{3}{5} * \frac{2}{5}$$

$$= \frac{3}{35} = 0.08$$

Ans

Pr(tes | Sunny, Hot)

$$= \frac{0.031}{(0.031 + 0.085)} = 0.27 = 27\%$$

Pr(no | Sunny, Hot)

$$= \frac{0.085}{(0.031 + 0.085)} = 0.73 = 73\%$$

Outlier	Temperature	O/P
Sunny	Hot	73% = They will not play tennis

27% ⇒
They will play tennis

(11)

0 = 1 person is not going
to play tennis

⇒ Variants of Naive Bayes

(1) Bernoulli Naive Bayes

(2) ~~Real~~ Multinomial Naive Bayes

(3) Gaussian Naive Bayes

(1) Bernoulli Naive Bayes:-

Whenever your features are following a Bernoulli Distribution, then we have to use Bernoulli Algorithms.

Reset

Serialli $\rightarrow$ $\frac{12}{0.11}$

b₁ b₂ b₃ O/P

Reset

b ₁	b ₂	b ₃	O/P
tes	res	male	tes
tes	Fail	Female	No
No	res	male	tes
tes	Fail	Female	No

(2) Multinomial Naive Bayes :-

Whenever, we have input data in form of text.

$\rightarrow$ we use Multinomial Naive Bayes

Refers :- spam classifi
cation

Spam :- Spam / not
spam

We receive
million \$\$
letters => Spam

First, you have
done good job :- HAM



Numerical
value => Natural
language
processing

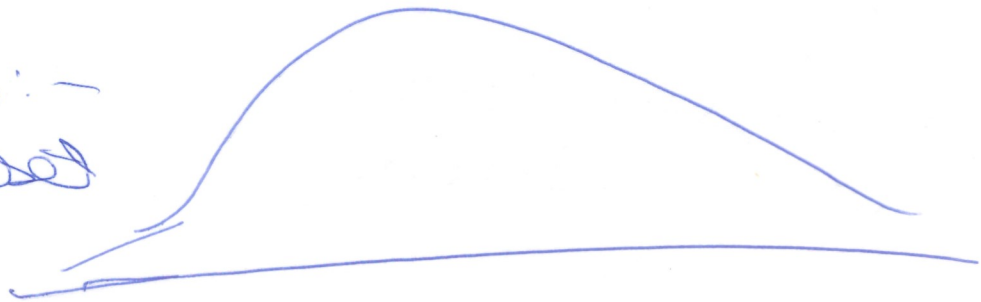
(1) Bow

(2) T f - T f f

(3) Gaussian, Naive Bayes :-

If the features are following Gaussian Distribution then, we use Gaussian Naive Bayes.

IRIS:-
~~Dataset~~



Dataset $\Rightarrow$ continuous

Age	Height	Weight	to/na
25	170	78	✓
	160	75	
38	150	60	✓
12	140	35	

where we have continuous features

(13)

⇒ For Name Based system
algorithm, the
whole data is mainly
converted to sparse
matrix.



Further used for
NLP problem statements